

Step 1

```
CREATE DATABASE DataAssignment;
```

```
use DataAssignment;
```

```
--Creating artist table
```

```
CREATE TABLE Artist (  
    ArtistID INT PRIMARY KEY IDENTITY(1,1),  
    Name VARCHAR(100) NOT NULL,  
    Surname VARCHAR(100) NOT NULL  
);
```

```
--Creating genre table
```

```
CREATE TABLE Genre(  
    GenreID INT PRIMARY KEY IDENTITY(1,1),  
    Description VARCHAR(255) NOT NULL  
);
```

```
--Creating artwork table
```

```
CREATE TABLE Artwork(  
    ArtworkID INT PRIMARY KEY IDENTITY(1,1),  
    Title VARCHAR(255) NOT NULL,  
    GenreID INT,  
    ArtistID INT,  
    FOREIGN KEY (GenreID) REFERENCES Genre(GenreID),  
    FOREIGN KEY (ArtistID) REFERENCES Artist(ArtistID)  
);
```

--Creating exhibition table

CREATE TABLE Exhibition(

ExhibitionID INT PRIMARY KEY IDENTITY(1,1),

Description VARCHAR(255) NOT NULL

);

--Creating entry table

CREATE TABLE Entry(

EntryID INT PRIMARY KEY IDENTITY(1,1),

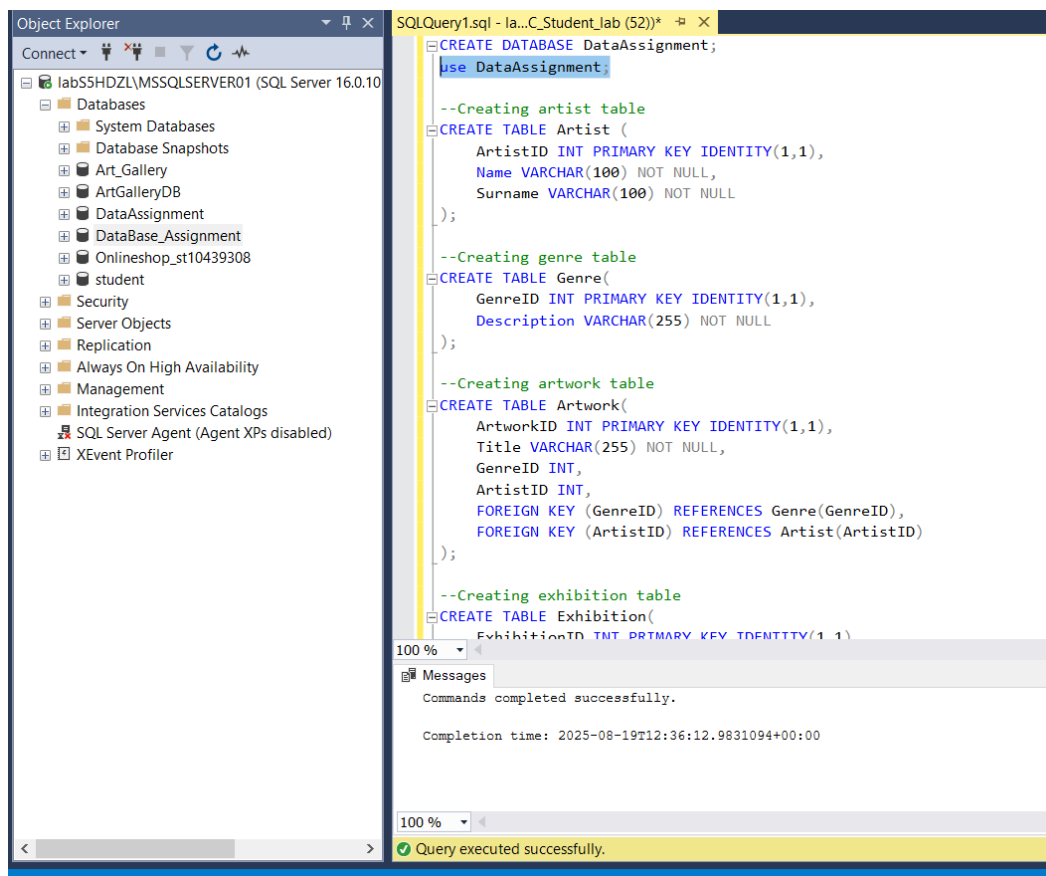
ArtworkID INT,

ExhibitionID INT,

FOREIGN KEY (ArtworkID) REFERENCES Artwork(ArtworkID),

FOREIGN KEY (ExhibitionID) REFERENCES Exhibition(ExhibitionID)

);



```
SELECT * From Artist;

SELECT * From Genre;

SELECT * FROM Artwork;

SELECT * FROM Exhibition;

SELECT * FROM Entry;
```

Step 2

--Insert Artist

```
INSERT INTO Artist (Name, Surname) VALUES

('Pablo','Picasso'),

('Vincent','Van Gogh'),

('Leonardo','da Vinci'),

('Claude','Monet'),

('Frida','Kahlo');
```

--Insert Genre

```
INSERT INTO Genre (Description) VALUES

('Abstract'),

('Impression'),

('Renasissance');
```

-- Insert Artworks

```
INSERT INTO Artwork (Title, GenreID, ArtistID) VALUES

('Guernica', 1, 1),

('Starry Night', 2, 2),

('Mona Lisa', 3, 3),

('Water Lilies', 2, 4),

('The Two Fridas', 1, 5),
```

('The Weeping Woman', 1, 1),
('Sunflowers', 2, 2),
('The Last Supper', 3, 3),
('Impression, Sunrise', 2, 4),
('Self Portrait with Thorn Necklace', 1, 5),
('Les Demoiselles d'Avignon', 1, 1),
('Irises', 2, 2),
('Vitruvian Man', 3, 3),
('Woman with a Parasol', 2, 4),
('Roots', 1, 5),
('La Vie', 1, 1),
('Bedroom in Arles', 2, 2),
('Annunciation', 3, 3),
('Haystacks', 2, 4),
('The Broken Column', 1, 5);

-- Insert Exhibitions

INSERT INTO Exhibition (Description) VALUES

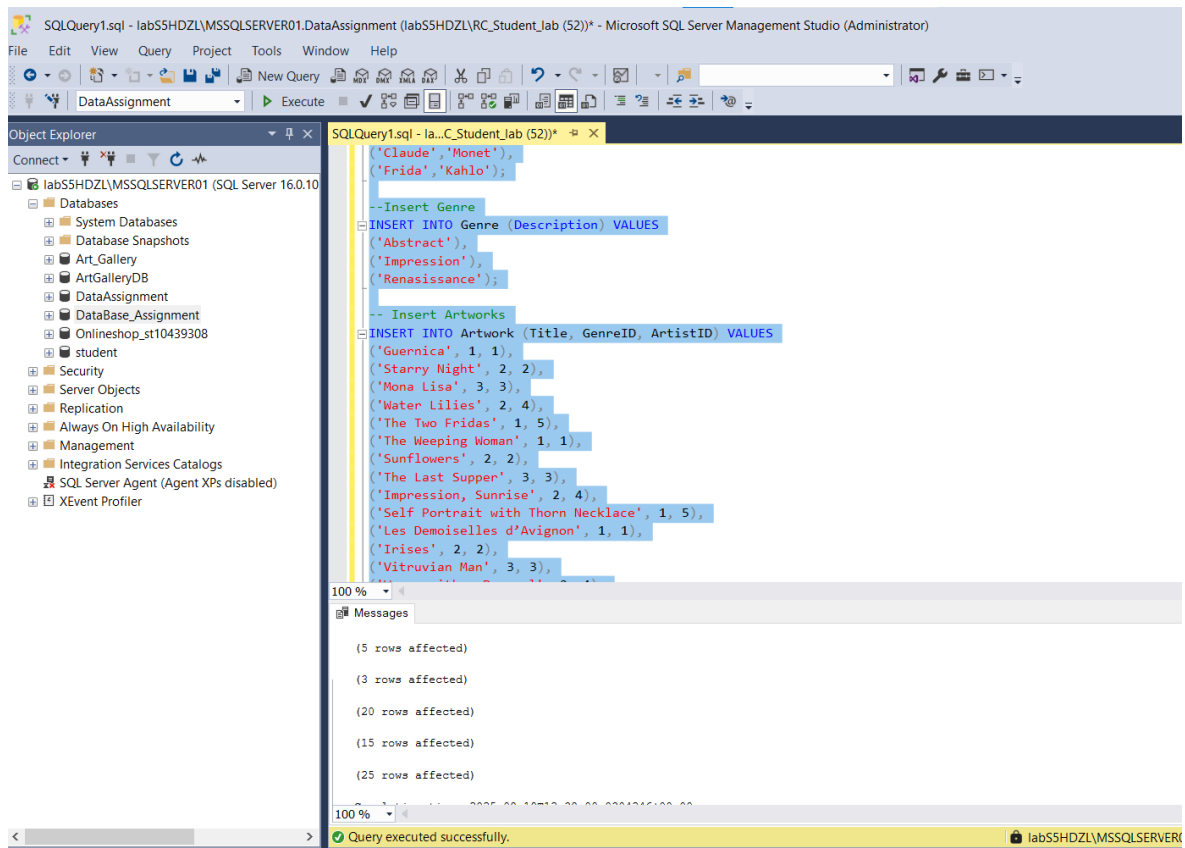
('Modern Art Showcase'),
('Impressionist Masters'),
('Renaissance Wonders'),
('Women in Art'),
('Masterpieces of Picasso'),
('Van Gogh Tribute'),
('Da Vinci Collection'),
('Monet Special'),
('Frida Kahlo's Legacy'),
('Abstract World'),
('Art Through the Ages'),

('Colors of Nature'),
('Classic to Modern'),
('Legends of Art'),
('Iconic Artworks');

-- Insert Entries (Artwork and Exhibition)

INSERT INTO Entry (ArtworkID, ExhibitionID) VALUES

(1, 1), (1, 5),
(2, 2), (2, 6),
(3, 3), (3, 7),
(4, 2), (4, 8),
(5, 4), (5, 9),
(6, 5),
(7, 6),
(8, 7),
(9, 2),
(10, 4),
(11, 10),
(12, 6),
(13, 7),
(14, 8),
(15, 9),
(16, 1),
(17, 6),
(18, 3),
(19, 8),
(20, 4);



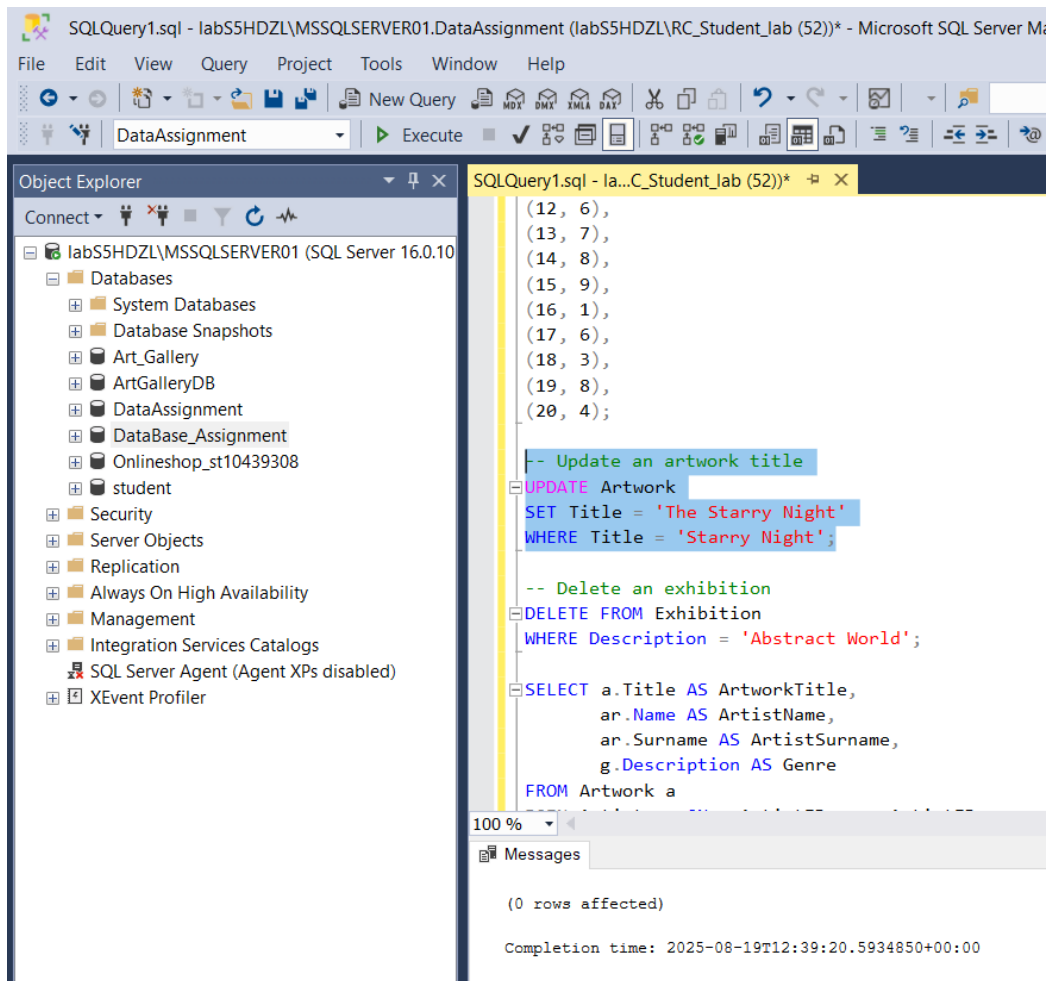
Step 3

-- Update an artwork title

UPDATE Artwork

SET Title = 'The Starry Night'

WHERE Title = 'Starry Night';

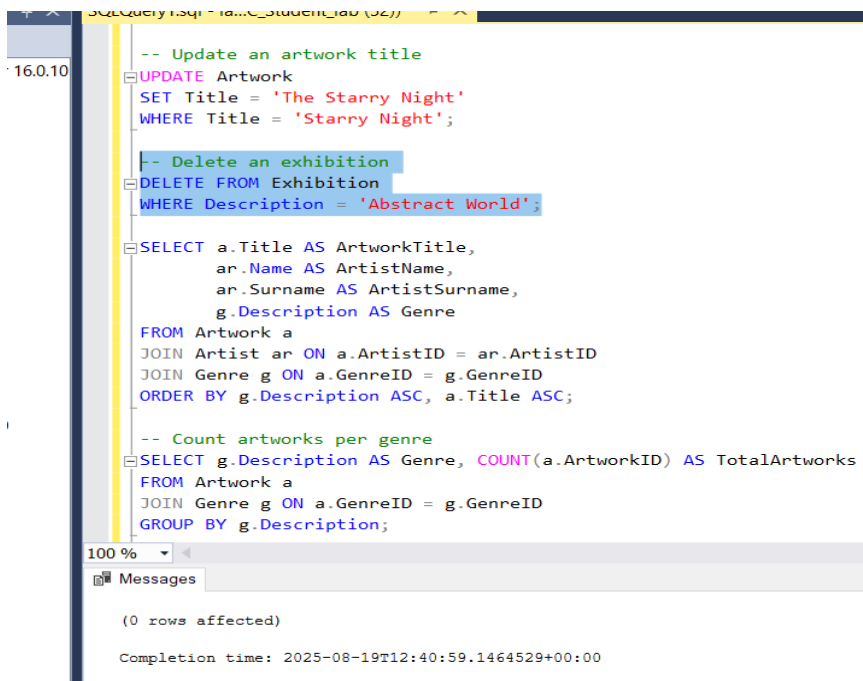


Step 4

-- Delete an exhibition

DELETE FROM Exhibition

WHERE Description = 'Abstract World';



Step 5

```
SELECT a.Title AS ArtworkTitle,  
       ar.Name AS ArtistName,  
       ar.Surname AS ArtistSurname,  
       g.Description AS Genre  
FROM Artwork a  
JOIN Artist ar ON a.ArtistID = ar.ArtistID  
JOIN Genre g ON a.GenreID = g.GenreID  
ORDER BY g.Description ASC, a.Title ASC;
```

The screenshot shows a SQL query editor with three queries. The first query is highlighted in blue. The second and third queries are in green. The results of the first query are shown in a table below the queries. The table has five columns: ArtworkTitle, ArtistName, ArtistSurname, and Genre. The results show 10 rows of data, including 'Guernica' by Pablo Picasso and 'Self Portrait with Thorn Necklace' by Frida Kahlo.

```
SQLQuery1.sql - la...C_Student_lab (52))* X  
16.0.10  
SELECT a.Title AS ArtworkTitle,  
       ar.Name AS ArtistName,  
       ar.Surname AS ArtistSurname,  
       g.Description AS Genre  
FROM Artwork a  
JOIN Artist ar ON a.ArtistID = ar.ArtistID  
JOIN Genre g ON a.GenreID = g.GenreID  
ORDER BY g.Description ASC, a.Title ASC;  
  
-- Count artworks per genre  
SELECT g.Description AS Genre, COUNT(a.ArtworkID) AS Tot  
FROM Artwork a  
JOIN Genre g ON a.GenreID = g.GenreID  
GROUP BY g.Description;  
  
-- Display genres that have more than 5 artworks  
SELECT g.Description AS Genre, COUNT(a.ArtworkID) AS Tot  
FROM Artwork a  
JOIN Genre g ON a.GenreID = g.GenreID  
GROUP BY g.Description  
HAVING COUNT(a.ArtworkID) > 5;  
  
-- List exhibitions with the artworks and their artists
```

	ArtworkTitle	ArtistName	ArtistSurname	Genre
1	Guernica	Pablo	Picasso	Abstract
2	Guernica	Pablo	Picasso	Abstract
3	La Vie	Pablo	Picasso	Abstract
4	La Vie	Pablo	Picasso	Abstract
5	Les Demoiselles d'Avignon	Pablo	Picasso	Abstract
6	Les Demoiselles d'Avignon	Pablo	Picasso	Abstract
7	Roots	Frida	Kahlo	Abstract
8	Roots	Frida	Kahlo	Abstract
9	Self Portrait with Thorn Necklace	Frida	Kahlo	Abstract
10	Self Portrait with Thorn Necklace	Frida	Kahlo	Abstract

100 %
Results Messages
Query executed successfully.

Step 6

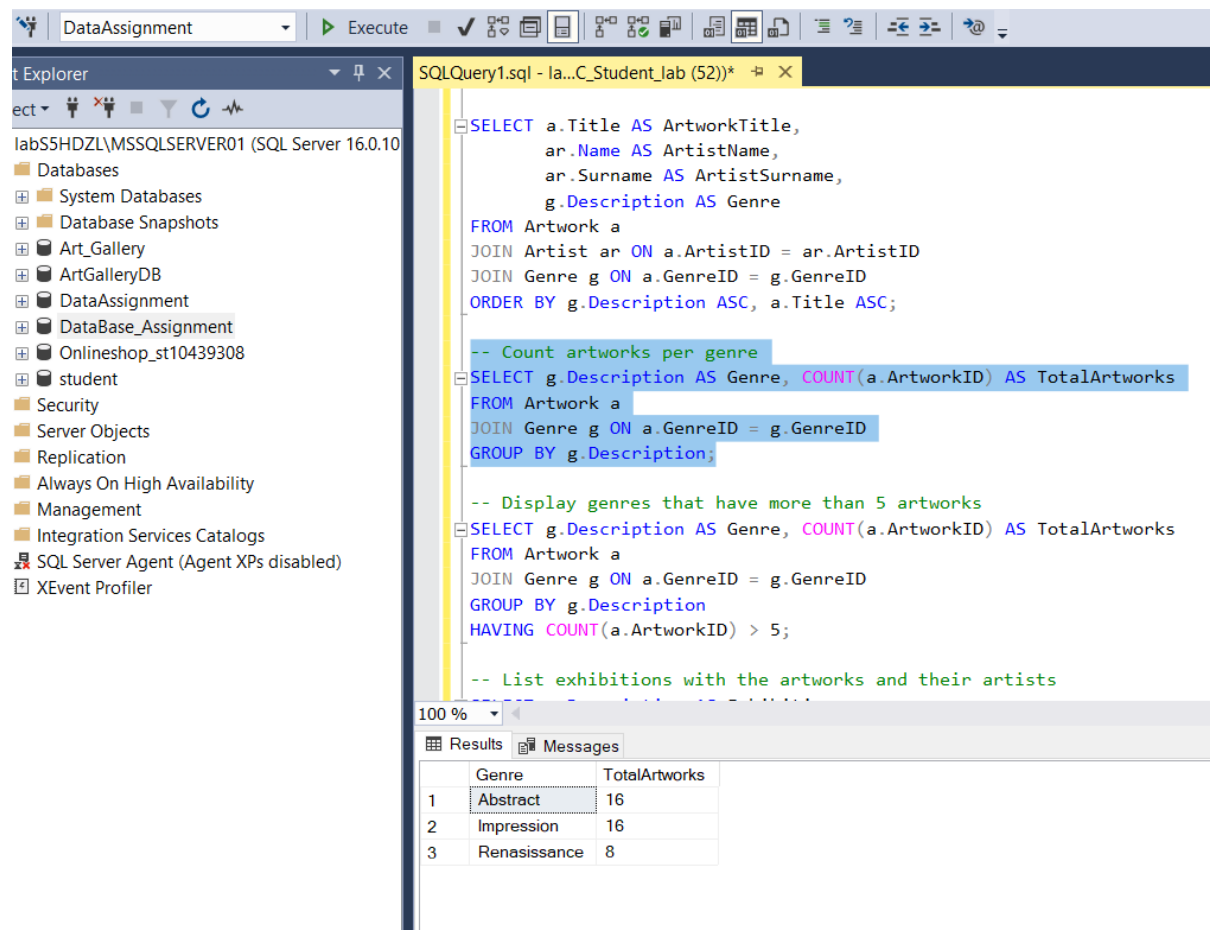
-- Count artworks per genre

```
SELECT g.Description AS Genre, COUNT(a.ArtworkID) AS TotalArtworks
```

```
FROM Artwork a
```

```
JOIN Genre g ON a.GenreID = g.GenreID
```

```
GROUP BY g.Description;
```



The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the server hierarchy for 'lab55HDZL\MSSQLSERVER01 (SQL Server 16.0.10)'. The right pane shows a query window titled 'SQLQuery1.sql - la...C_Student_lab (52)*'. The query contains three SQL statements: a query to select artwork details, a query to count artworks per genre, and a query to display genres with more than 5 artworks. The results window at the bottom shows the output of the third query, displaying a table with 3 rows and 2 columns: Genre and TotalArtworks.

```
-- Count artworks per genre
SELECT g.Description AS Genre, COUNT(a.ArtworkID) AS TotalArtworks
FROM Artwork a
JOIN Genre g ON a.GenreID = g.GenreID
GROUP BY g.Description;
```

```
-- Display genres that have more than 5 artworks
SELECT g.Description AS Genre, COUNT(a.ArtworkID) AS TotalArtworks
FROM Artwork a
JOIN Genre g ON a.GenreID = g.GenreID
GROUP BY g.Description
HAVING COUNT(a.ArtworkID) > 5;
```

	Genre	TotalArtworks
1	Abstract	16
2	Impression	16
3	Renasissance	8

Step 7

-- Display genres that have more than 5 artworks

```
SELECT g.Description AS Genre, COUNT(a.ArtworkID) AS TotalArtworks
```

```
FROM Artwork a
```

```
JOIN Genre g ON a.GenreID = g.GenreID
```

```
GROUP BY g.Description
```

```
HAVING COUNT(a.ArtworkID) > 5;
```

The screenshot shows a SQL Server Enterprise Manager window with a query editor and a results pane. The query editor contains the following SQL code:

```

JOIN Artist ar ON a.ArtistID = ar.ArtistID
JOIN Genre g ON a.GenreID = g.GenreID
ORDER BY g.Description ASC, a.Title ASC;

-- Count artworks per genre
SELECT g.Description AS Genre, COUNT(a.ArtworkID) AS TotalArtworks
FROM Artwork a
JOIN Genre g ON a.GenreID = g.GenreID
GROUP BY g.Description;

-- Display genres that have more than 5 artworks
SELECT g.Description AS Genre, COUNT(a.ArtworkID) AS TotalArtworks
FROM Artwork a
JOIN Genre g ON a.GenreID = g.GenreID
GROUP BY g.Description
HAVING COUNT(a.ArtworkID) > 5;

-- List exhibitions with the artworks and their artists
SELECT e.Description AS Exhibition,
       a.Title AS Artwork,
       ar.Name AS ArtistName,
       ar.Surname AS ArtistSurname
FROM Entry en
JOIN Exhibition e ON en.ExhibitionID = e.ExhibitionID

```

The results pane shows the output of the last query, which lists exhibitions with their artworks and artists. The results are as follows:

	Genre	TotalArtworks
1	Abstract	16
2	Impression	16
3	Renaissance	8

The status bar at the bottom indicates "Query executed successfully." and "Ln 150".

Step 8

-- List exhibitions with the artworks and their artists

SELECT e.Description AS Exhibition,

a.Title AS Artwork,

ar.Name AS ArtistName,

ar.Surname AS ArtistSurname

FROM Entry en

JOIN Exhibition e ON en.ExhibitionID = e.ExhibitionID

JOIN Artwork a ON en.ArtworkID = a.ArtworkID

JOIN Artist ar ON a.ArtistID = ar.ArtistID

ORDER BY e.Description;

DataAssignment Execute [Icons]

Object Explorer SQLQuery1.sql - la...C_Student_lab (52)*

Connect [Icons]

labSSHDLZ\MSSQLSERVER01 (SQL Server 16.0.10)

- Databases
 - System Databases
 - Database Snapshots
 - Art_Gallery
 - ArtGalleryDB
 - DataAssignment
 - DataBase_Assignment
 - Onlineshop_st10439308
 - student
- Security
- Server Objects
- Replication
- Always On High Availability
- Management
- Integration Services Catalogs
- SQL Server Agent (Agent XPs disabled)
- XEvent Profiler

```

FROM Artwork a
JOIN Genre g ON a.GenreID = g.GenreID
GROUP BY g.Description;

-- Display genres that have more than 5 artworks
SELECT g.Description AS Genre, COUNT(a.ArtworkID) AS TotalA
FROM Artwork a
JOIN Genre g ON a.GenreID = g.GenreID
GROUP BY g.Description
HAVING COUNT(a.ArtworkID) > 5;

-- List exhibitions with the artworks and their artists
SELECT e.Description AS Exhibition,
       a.Title AS Artwork,
       ar.Name AS ArtistName,
       ar.Surname AS ArtistSurname
FROM Entry en
JOIN Exhibition e ON en.ExhibitionID = e.ExhibitionID
JOIN Artwork a ON en.ArtworkID = a.ArtworkID
JOIN Artist ar ON a.ArtistID = ar.ArtistID
ORDER BY e.Description;

```

100 %

Results Messages

	Exhibition	Artwork	ArtistName	ArtistSurname
1	Abstract World	Les Demoiselles d'Avignon	Pablo	Picasso
2	Abstract World	Les Demoiselles d'Avignon	Pablo	Picasso
3	Da Vinci Collection	The Last Supper	Leonardo	da Vinci
4	Da Vinci Collection	Vitruvian Man	Leonardo	da Vinci
5	Da Vinci Collection	Vitruvian Man	Leonardo	da Vinci
6	Da Vinci Collection	The Last Supper	Leonardo	da Vinci
7	Da Vinci Collection	Mona Lisa	Leonardo	da Vinci
8	Da Vinci Collection	Mona Lisa	Leonardo	da Vinci
9	Frida Kahlo's Legacy	The Two Fridas	Frida	Kahlo
10	Frida Kahlo's Legacy	Roots	Frida	Kahlo

Query executed successfully.